Group: 2405
Section: 24

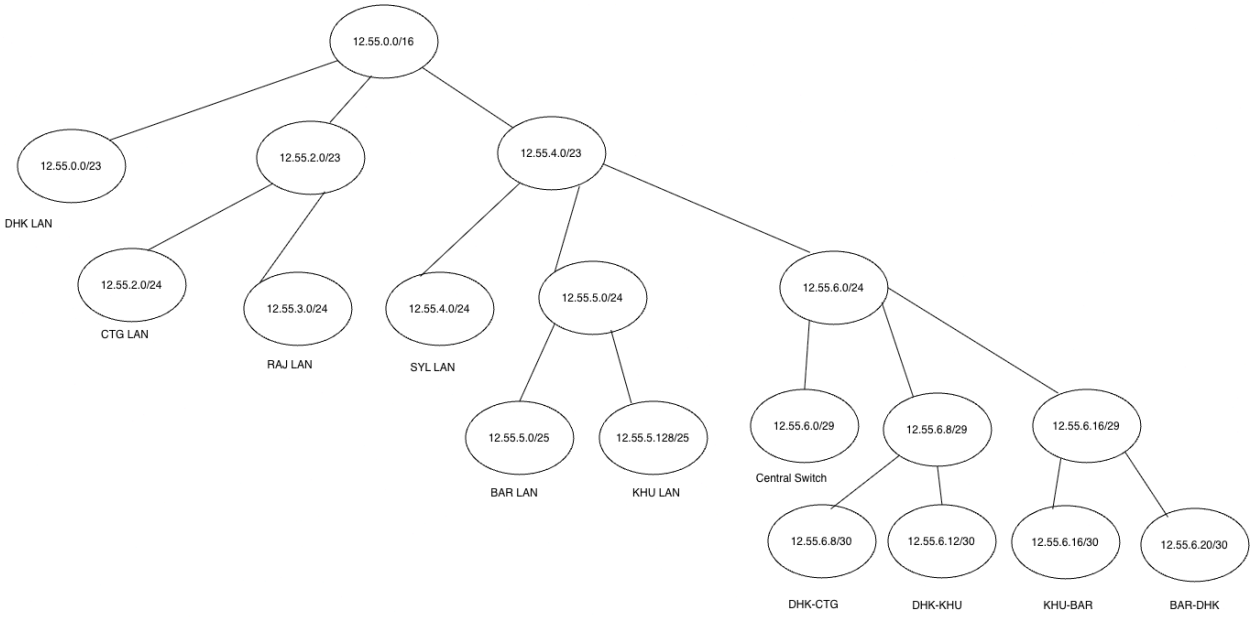
Topic 7: BoardNet: Board Examination Communication Network

1. Complete VLSM Tree:

Last 4 Digit: 1255
Split: 12 and 55
Base Network: 12.55.0.0/16

Number of hosts	Network Name	Block Size	Network Address	Usable Range	Broadcast	Subnet Mask
300	DHK LAN	512	12.55.0.0/23	.0.1 – .1.254	12.55.1.255	255.255.254.0
200	CTG LAN	256	12.55.2.0/24	.2.1 – .2.254	12.55.2.255	255.255.255.0
160	RAJ LAN	256	12.55.3.0/24	.3.1 – .3.254	12.55.3.255	255.255.255.0
150	SYL LAN	256	12.55.4.0/24	.4.1 – .4.254	12.55.4.255	255.255.255.0
120	BAR LAN	128	12.55.5.0/24	.5.1 – .5.126	12.55.5.127	255.255.255.128
120	KHU LAN	128	12.55.5.128/25	.5.129 – .5.254	12.55.5.255	255.255.255.128
3	Central switch	8	12.55.6.0/29	.6.1 – .6.6	12.55.6.7	255.255.255.248
2	DHK–CTG	4	12.55.6.8/30	.6.9 – .6.10	12.55.6.11	255.255.255.252
2	DHK–KHU	4	12.55.6.12/30	.6.13 – .6.14	12.55.6.15	255.255.255.252
2	KHU–BAR	4	12.55.6.16/30	.6.17 – .6.18	12.55.6.19	255.255.255.252
2	BAR–DHK	4	12.55.6.20/30	.6.21 – .6.22	12.55.6.23	255.255.255.252

VLSM Tree


12.55.0.0/16
12.55.0.0/23
DHK LAN
12.55.2.0/23
12.55.4.0/23
12.55.2.0/24
CTG LAN
12.55.3.0/24
RAJ LAN
12.55.4.0/24
SYL LAN
12.55.5.0/24
12.55.6.0/24
12.55.5.0/25
BAR LAN
12.55.5.128/25
KHU LAN
12.55.6.0/29
Central Switch
12.55.6.8/29
12.55.6.16/29
12.55.6.8/30
DHK-CTG
12.55.6.12/30
DHK-KHU
12.55.6.16/30
KHU-BAR
12.55.6.20/30
BAR-DHK

2. **Full IP addressing table (unit, device, IP address, subnet mask, default gateway)**

DHK Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
DHK	R-DHK	G0/0	12.55.0.1	255.255.254.0	—
		G0/1	12.55.6.1	255.255.255.248	—
		S0/0/0 (DCE→CTG)	12.55.6.9	255.255.255.252	—
		S0/0/1 (DCE→KHU)	12.55.6.13	255.255.255.252	—
		S0/1/0 (DCE→BAR)	12.55.6.22	255.255.255.252	—
	DNS-Server	NIC	12.55.0.2	255.255.254.0	12.55.0.1
	WEB-DHK	NIC	12.55.0.3	255.255.254.0	12.55.0.1
	MAIL-DHK	NIC	12.55.0.4	255.255.254.0	12.55.0.1
	PRN-DHK	NIC	12.55.0.5	255.255.254.0	12.55.0.1
	PC-DHK1/2	NIC	DHCP	DHCP	DHCP

CTG Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
CTG	R-CTG	G0/0	12.55.2.1	255.255.255.0	—
		S0/0/0 (DTE→DHK)	12.55.6.10	255.255.255.252	—
	WEB-CTG	NIC	12.55.2.3	255.255.255.0	12.55.2.1
	MAIL-CTG	NIC	12.55.2.4	255.255.255.0	12.55.2.1
	PRN-CTG	NIC	12.55.2.5	255.255.255.0	12.55.2.1
	PC-CTG1/2	NIC	DHCP	DHCP	DHCP

RAJ Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
RAJ	R-RAJ	G0/0	12.55.3.1	255.255.255.0	—
		G0/1 (→central SW)	12.55.6.3	255.255.255.248	—
	MAIL-RAJ	NIC	12.55.3.4	255.255.255.0	12.55.3.1
	PRN-RAJ	NIC	12.55.3.5	255.255.255.0	12.55.3.1
	PC-RAJ1/2	NIC	DHCP	DHCP	DHCP

SYL Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
SYL	R-SYL	G0/0	12.55.4.1	255.255.255.0	—
		G0/1 (→central SW)	12.55.6.2	255.255.255.248	—
	MAIL-SYL	NIC	12.55.4.4	255.255.255.0	12.55.4.1
	PRN-SYL	NIC	12.55.4.5	255.255.255.0	12.55.4.1
	PC-SYL1/2	NIC	DHCP	DHCP	DHCP

BAR Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
BAR	R-BAR	G0/0	12.55.5.1	255.255.255.128	—
		S0/0/0 (DTE→KHU)	12.55.6.18	255.255.255.252	—
		S0/0/1 (DTE→DHK)	12.55.6.21	255.255.255.252	—
	DHCP-BAR	NIC	12.55.5.2	255.255.255.128	12.55.5.1
	MAIL-BAR	NIC	12.55.5.3	255.255.255.128	12.55.5.1
	PRN-BAR	NIC	12.55.5.4	255.255.255.128	12.55.5.1
	PC-BAR1/2	NIC	DHCP	DHCP	DHCP

KHU Board:

Unit	Device	Interface	IP Address	Subnet Mask	Default Gateway
KHU	R-KHU	G0/0	12.55.5.129	255.255.255.128	—
		S0/0/0 (DTE→DHK)	12.55.6.14	255.255.255.252	—
		S0/0/1 (DCE→BAR)	12.55.6.17	255.255.255.252	—
	DHCP-KHU	NIC	12.55.5.130	255.255.255.128	12.55.5.129
	MAIL-KHU	NIC	12.55.5.131	255.255.255.128	12.55.5.129
	PRN-KHU	NIC	12.55.5.132	255.255.255.128	12.55.5.129
	PC-KHU1/2	NIC	DHCP	DHCP	DHCP

DNS server for every device = 12.55.0.2

3. The configuration commands of all the 6 routers we have implemented.

Router base configuration:

R-DHK	**R-CTG**	**R-RAJ**
enable configure terminal hostname R-DHK no ip domain-lookup enable secret class banner motd & ========== BoardNet - Authorized Access Only ========== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config	enable configure terminal hostname R-CTG no ip domain-lookup enable secret class banner motd & ========== BoardNet - Authorized Access Only ========== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config	enable configure terminal hostname R-RAJ no ip domain-lookup enable secret class banner motd & ========== BoardNet - Authorized Access Only ========== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config

R-SYL	**R-BAR**	**R-KHU**
enable configure terminal hostname R-SYL	enable configure terminal hostname R-BAR	enable configure terminal hostname R-KHU

no ip domain-lookup enable secret class banner motd & ====================== BoardNet - Authorized Access Only ====================== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config	no ip domain-lookup enable secret class banner motd & ====================== BoardNet - Authorized Access Only ====================== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config	no ip domain-lookup enable secret class banner motd & ====================== BoardNet - Authorized Access Only ====================== & line console 0 password cisco login exit line vty 0 4 password cisco login exit end copy running-config startup-config

R-DHK full configuration:

configure terminal
interface g0/0
ip address 12.55.0.1 255.255.254.0
no shutdown
exit
interface g0/1
ip address 12.55.6.1 255.255.255.248
no shutdown
exit
interface s0/0/0
ip address 12.55.6.9 255.255.255.252
clock rate 64000
no shutdown
exit
interface s0/0/1
ip address 12.55.6.13 255.255.255.252
clock rate 64000
no shutdown

```
exit
interface s0/1/0
 ip address 12.55.6.22 255.255.255.252
 clock rate 64000
 no shutdown
 end
```

On R-DHK — excluded addresses and the four pools:

```
configure terminal
ip dhcp excluded-address 12.55.0.1 12.55.0.10
ip dhcp excluded-address 12.55.2.1 12.55.2.10
ip dhcp excluded-address 12.55.3.1 12.55.3.10
ip dhcp excluded-address 12.55.4.1 12.55.4.10
ip dhcp pool DHK_LAN
 network 12.55.0.0 255.255.254.0
 default-router 12.55.0.1
 dns-server 12.55.0.2
 exit
ip dhcp pool CTG_LAN
 network 12.55.2.0 255.255.255.0
 default-router 12.55.2.1
 dns-server 12.55.0.2
 exit
ip dhcp pool RAJ_LAN
 network 12.55.3.0 255.255.255.0
 default-router 12.55.3.1
 dns-server 12.55.0.2
 exit
ip dhcp pool SYL_LAN
 network 12.55.4.0 255.255.255.0
 default-router 12.55.4.1
 dns-server 12.55.0.2
 end
```

---------- RIPv2 with SYL and RAJ ----------

```
router rip
version 2
no auto-summary
network 12.0.0.0
passive-interface GigabitEthernet0/0
passive-interface Serial0/0/0
passive-interface Serial0/0/1
passive-interface Serial0/1/0
```

default-information originate
exit

---------- static routes to the non-RIP boards ----------
ip route 12.55.2.0 255.255.255.0 12.55.6.10
ip route 12.55.5.0 255.255.255.128 12.55.6.21
ip route 12.55.5.128 255.255.255.128 12.55.6.14
ip route 12.55.6.16 255.255.255.252 12.55.6.14
ip route 12.55.5.0 255.255.255.128 12.55.6.14 5
ip route 12.55.5.128 255.255.255.128 12.55.6.21 5
end
copy running-config startup-config

R-CTG full configuration:

interface GigabitEthernet0/0
ip address 12.55.2.1 255.255.255.0
ip helper-address 12.55.6.9
no shutdown
interface Serial0/0/0
ip address 12.55.6.10 255.255.255.252
no shutdown
ip route 0.0.0.0 0.0.0.0 12.55.6.9
end
copy running-config startup-config

R-SYL full configuration:

interface GigabitEthernet0/0
ip address 12.55.4.1 255.255.255.0
ip helper-address 12.55.6.1
no shutdown
interface GigabitEthernet0/1
ip address 12.55.6.2 255.255.255.248
no shutdown

router rip
version 2
no auto-summary

```
network 12.0.0.0
passive-interface GigabitEthernet0/0
end
copy running-config startup-config
```

R-RAJ full configuration:

```
interface GigabitEthernet0/0
ip address 12.55.3.1 255.255.255.0
ip helper-address 12.55.6.1
no shutdown
interface GigabitEthernet0/1
ip address 12.55.6.3 255.255.255.248
no shutdown
router rip
version 2
no auto-summary
network 12.0.0.0
passive-interface GigabitEthernet0/0
end
copy running-config startup-config
```

R-KHU full configuration:

```
interface GigabitEthernet0/0
ip address 12.55.5.129 255.255.255.128
no shutdown

interface Serial0/0/0
ip address 12.55.6.14 255.255.255.252
no shutdown

interface Serial0/0/1
ip address 12.55.6.17 255.255.255.252
clock rate 64000
no shutdown
```

----- **primary recursive static routes via DHK (AD 1)** -----
ip route 12.55.0.0 255.255.254.0 12.55.6.13
ip route 12.55.2.0 255.255.255.0 12.55.6.13
ip route 12.55.3.0 255.255.255.0 12.55.6.13
ip route 12.55.4.0 255.255.255.0 12.55.6.13
ip route 12.55.6.0 255.255.255.248 12.55.6.13
ip route 12.55.6.8 255.255.255.252 12.55.6.13
ip route 12.55.6.20 255.255.255.252 12.55.6.13
ip route 12.55.5.0 255.255.255.128 12.55.6.18

----- **floating backups via BAR (AD 5)** -----
ip route 12.55.0.0 255.255.254.0 12.55.6.18 5
ip route 12.55.2.0 255.255.255.0 12.55.6.18 5
ip route 12.55.3.0 255.255.255.0 12.55.6.18 5
ip route 12.55.4.0 255.255.255.0 12.55.6.18 5
ip route 12.55.6.0 255.255.255.248 12.55.6.18 5
ip route 12.55.6.8 255.255.255.252 12.55.6.18 5
ip route 12.55.6.20 255.255.255.252 12.55.6.18 5
ip route 12.55.5.0 255.255.255.128 12.55.6.13 5
end
copy running-config startup-config

R-BAR full configuration:

interface GigabitEthernet0/0
ip address 12.55.5.1 255.255.255.128
no shutdown

interface Serial0/0/0
ip address 12.55.6.18 255.255.255.252
no shutdown

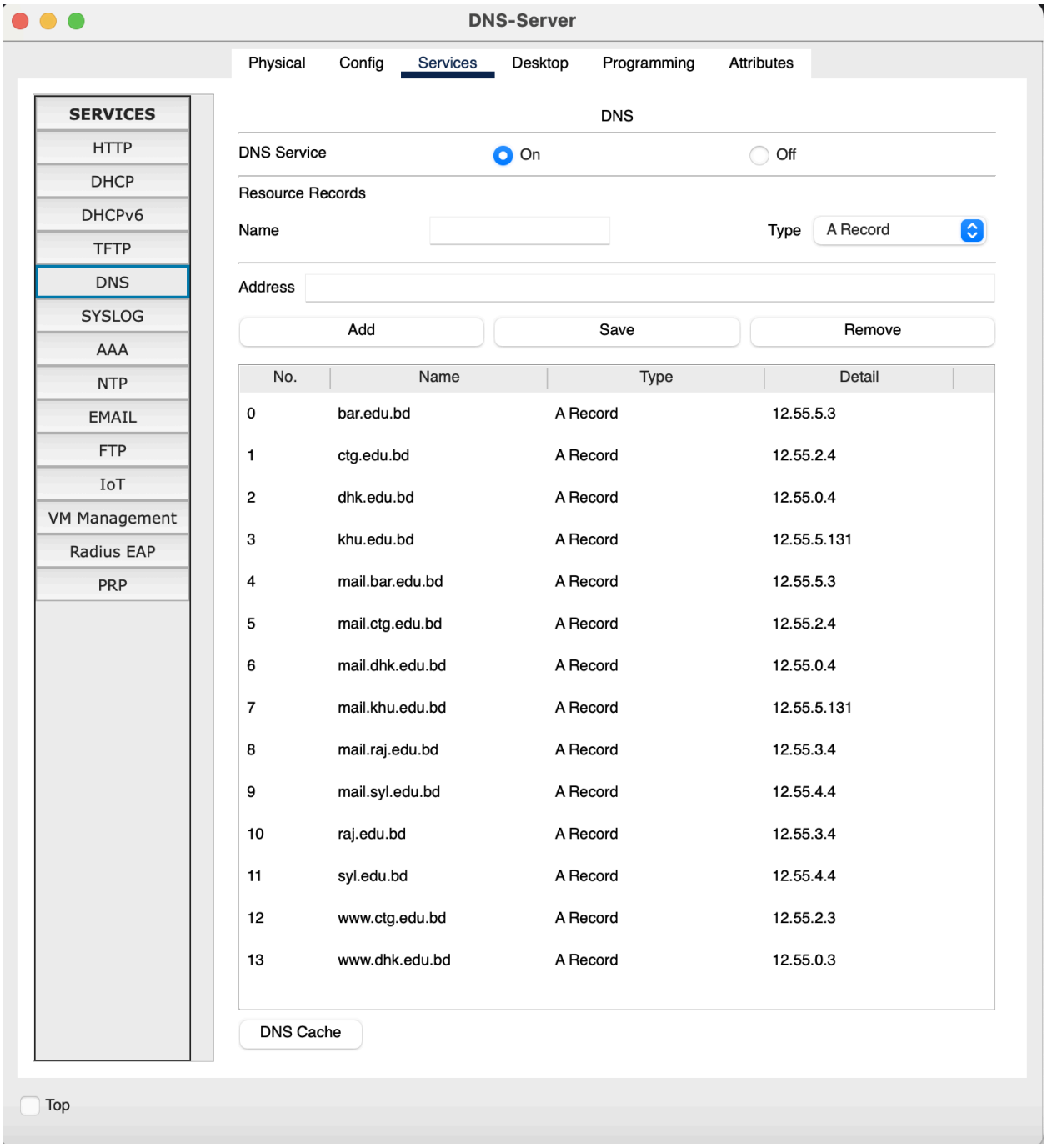
interface Serial0/0/1
ip address 12.55.6.21 255.255.255.252
no shutdown

----- **primary recursive static routes via DHK (AD 1)** -----

ip route 12.55.0.0 255.255.254.0 12.55.6.22
ip route 12.55.2.0 255.255.255.0 12.55.6.22
ip route 12.55.3.0 255.255.255.0 12.55.6.22
ip route 12.55.4.0 255.255.255.0 12.55.6.22
ip route 12.55.6.0 255.255.255.248 12.55.6.22
ip route 12.55.6.8 255.255.255.252 12.55.6.22
ip route 12.55.6.12 255.255.255.252 12.55.6.22
ip route 12.55.5.128 255.255.255.128 12.55.6.17

----- **floating backups via KHU (AD 5)** -----
ip route 12.55.0.0 255.255.254.0 12.55.6.17 5
ip route 12.55.2.0 255.255.255.0 12.55.6.17 5
ip route 12.55.3.0 255.255.255.0 12.55.6.17 5
ip route 12.55.4.0 255.255.255.0 12.55.6.17 5
ip route 12.55.6.0 255.255.255.248 12.55.6.17 5
ip route 12.55.6.8 255.255.255.252 12.55.6.17 5
ip route 12.55.6.12 255.255.255.252 12.55.6.17 5
ip route 12.55.5.128 255.255.255.128 12.55.6.22 5
end
copy running-config startup-config

4. DNS configuration:



5. DHCP configuration:

R-DHK :

```
R-DHK>enable
Password:
R-DHK#show ip dhcp pool

Pool DHK_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 2
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 12.55.0.1            12.55.0.1        - 12.55.1.254       2    / 4     / 510

Pool CTG_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 12.55.2.1            12.55.2.1        - 12.55.2.254       2    / 4     / 254

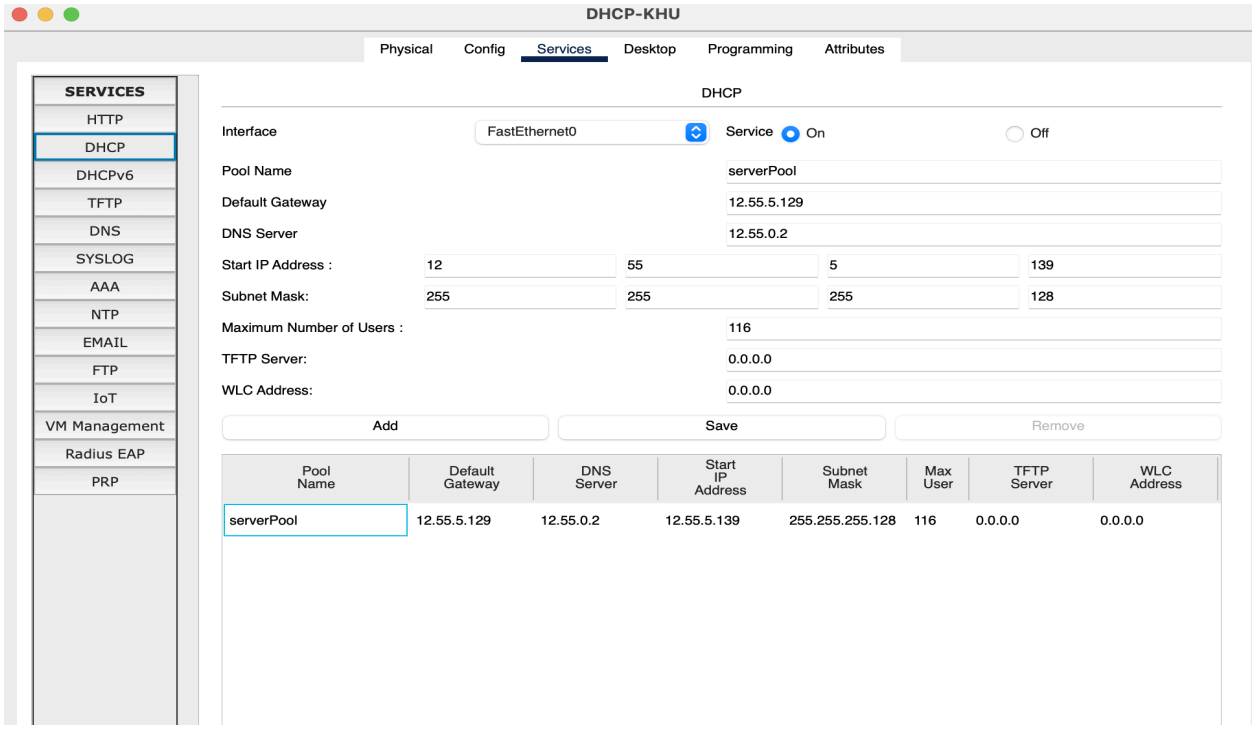
Pool RAJ_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 12.55.3.1            12.55.3.1        - 12.55.3.254       2    / 4     / 254

Pool SYL_LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 4
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 12.55.4.1            12.55.4.1        - 12.55.4.254       2    / 4     / 254
R-DHK#
```

DHCP-KHU :



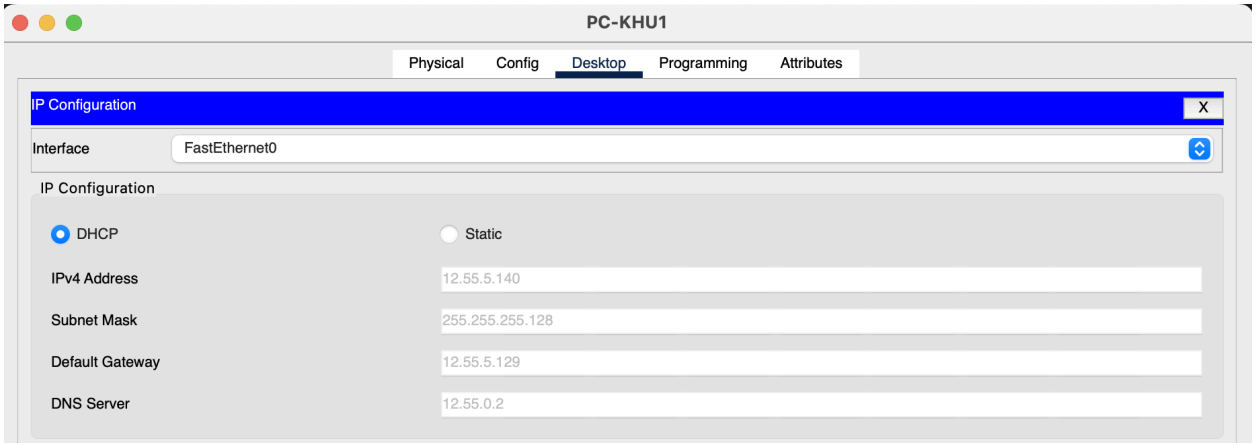
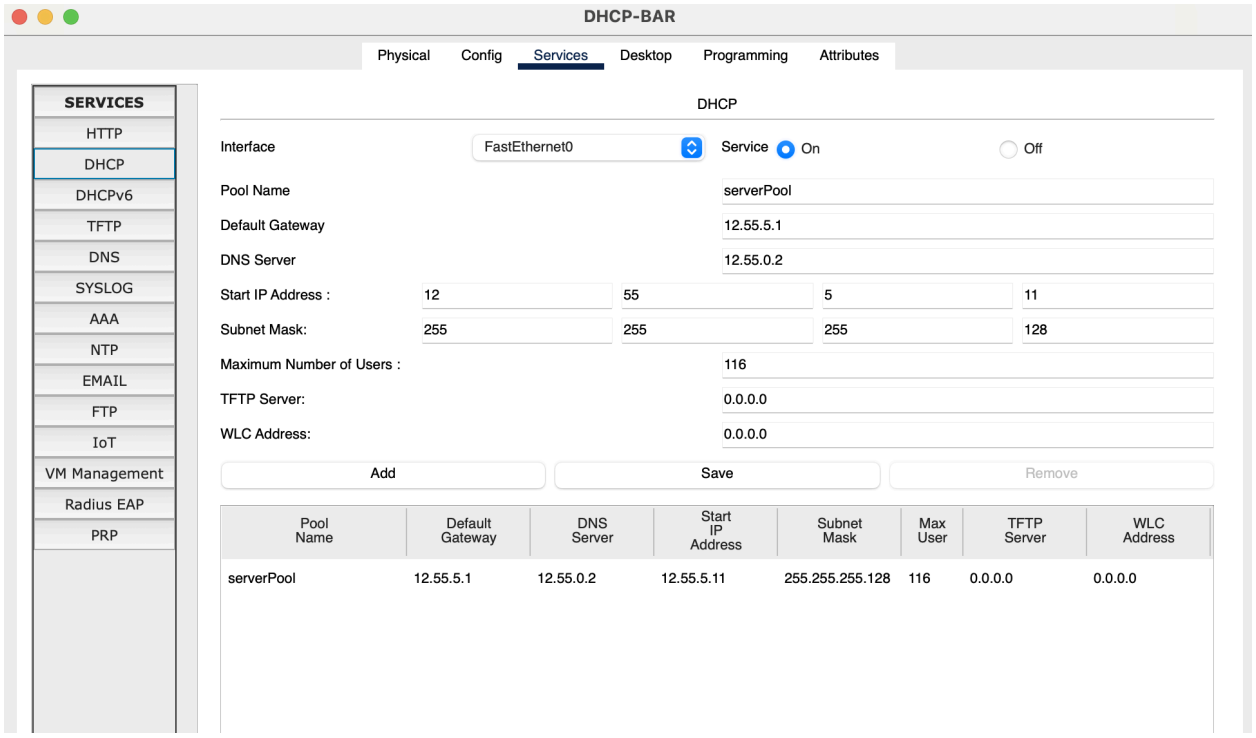


Fig: PC-KHU1 received an IP

DHCP-BAR :



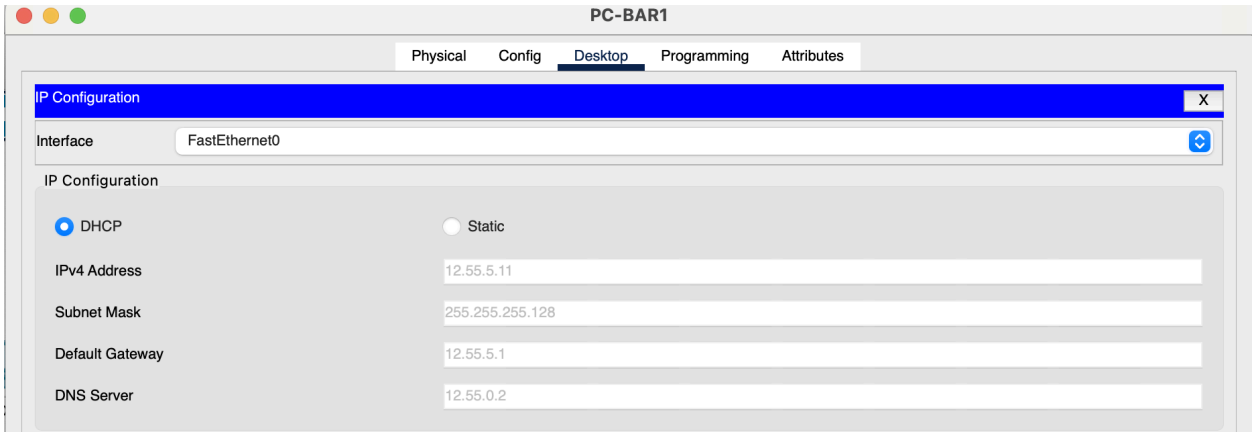
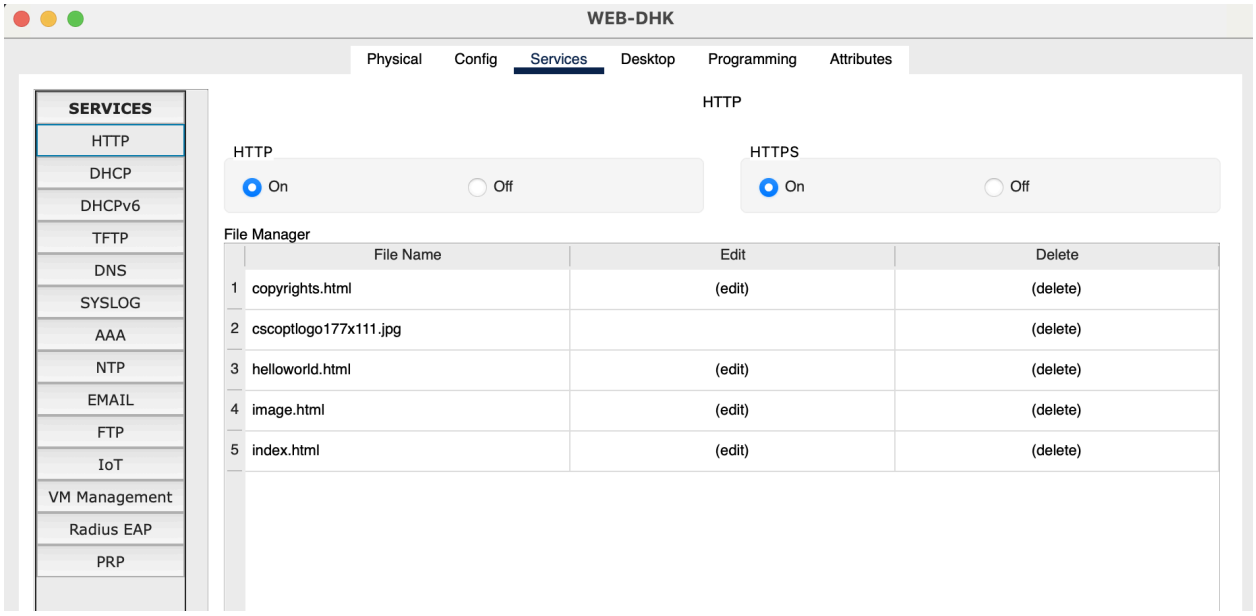


Fig: PC-BAR1 received an IP

6. Web server configuration details:

WEB-DHK:



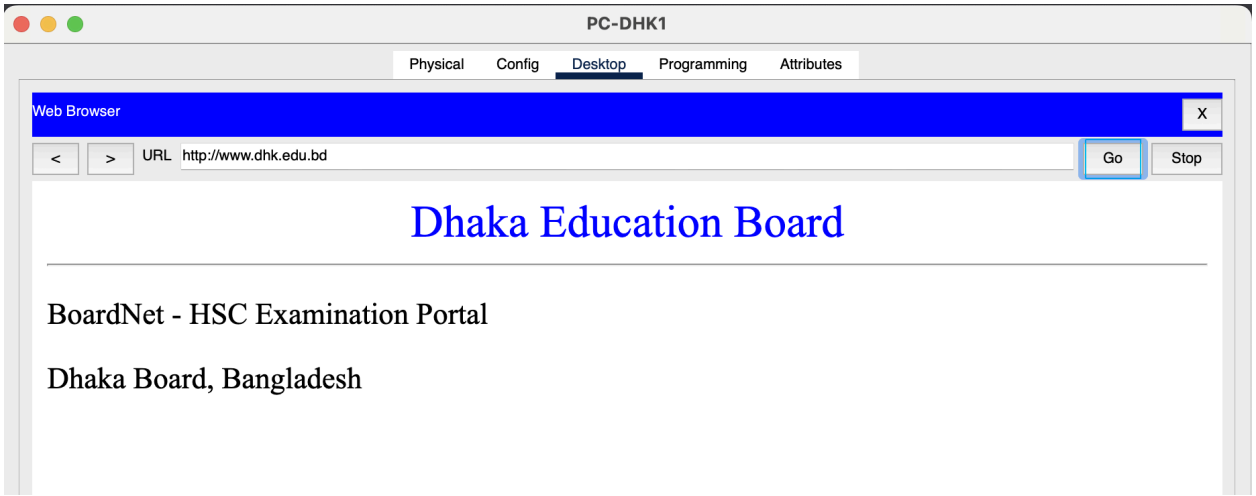
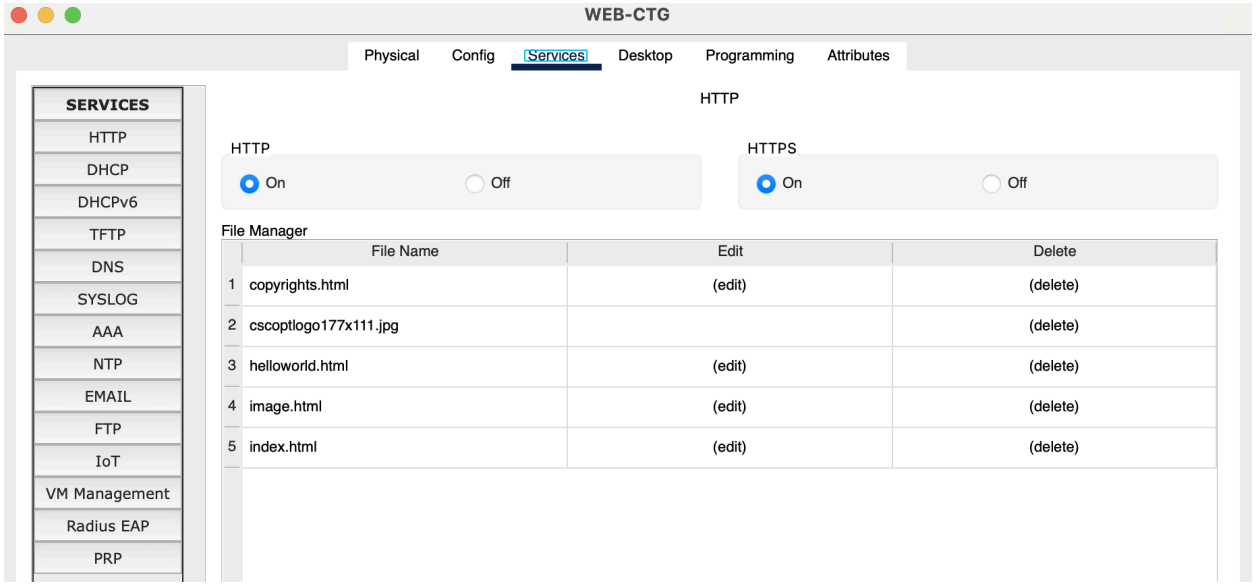


Fig: WEB-DHK can be accessed from PC-DHK1

WEB-CTG:



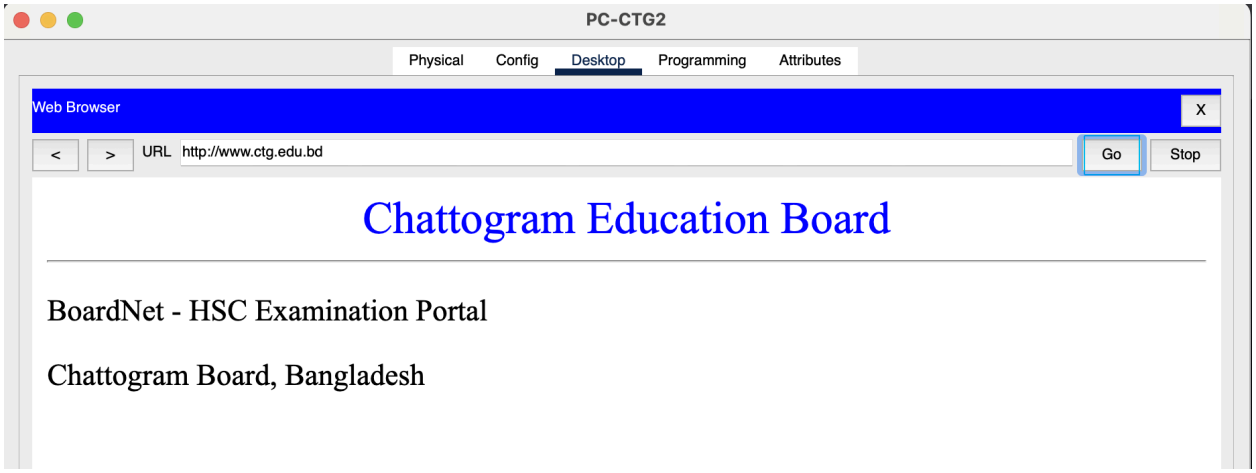
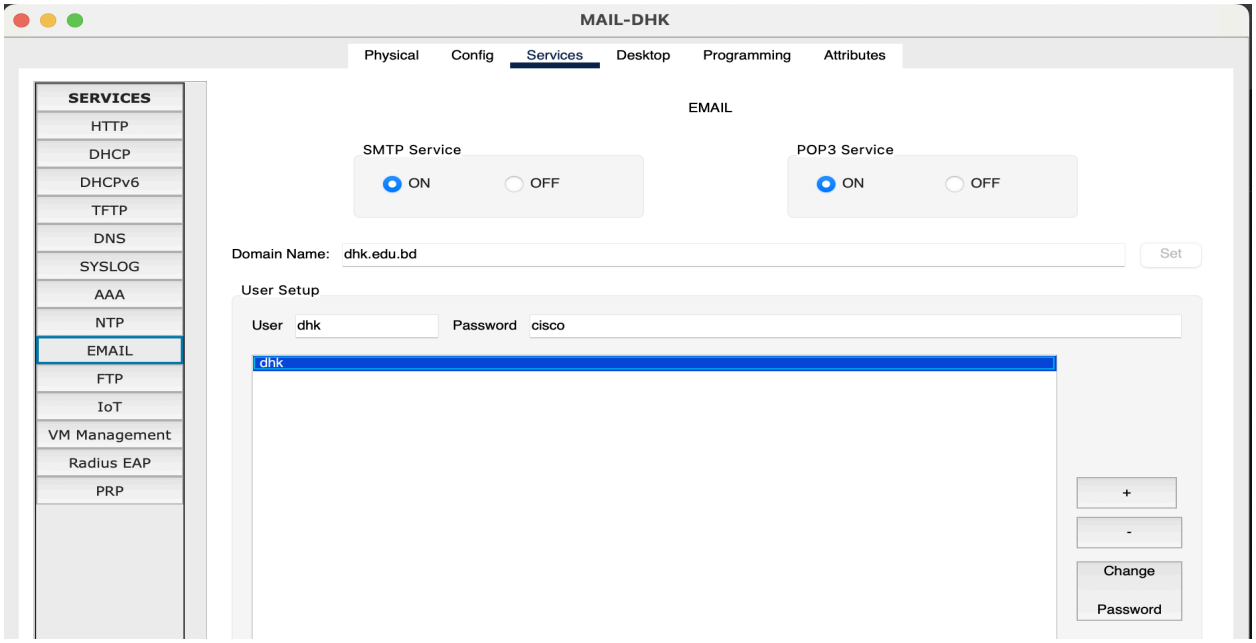


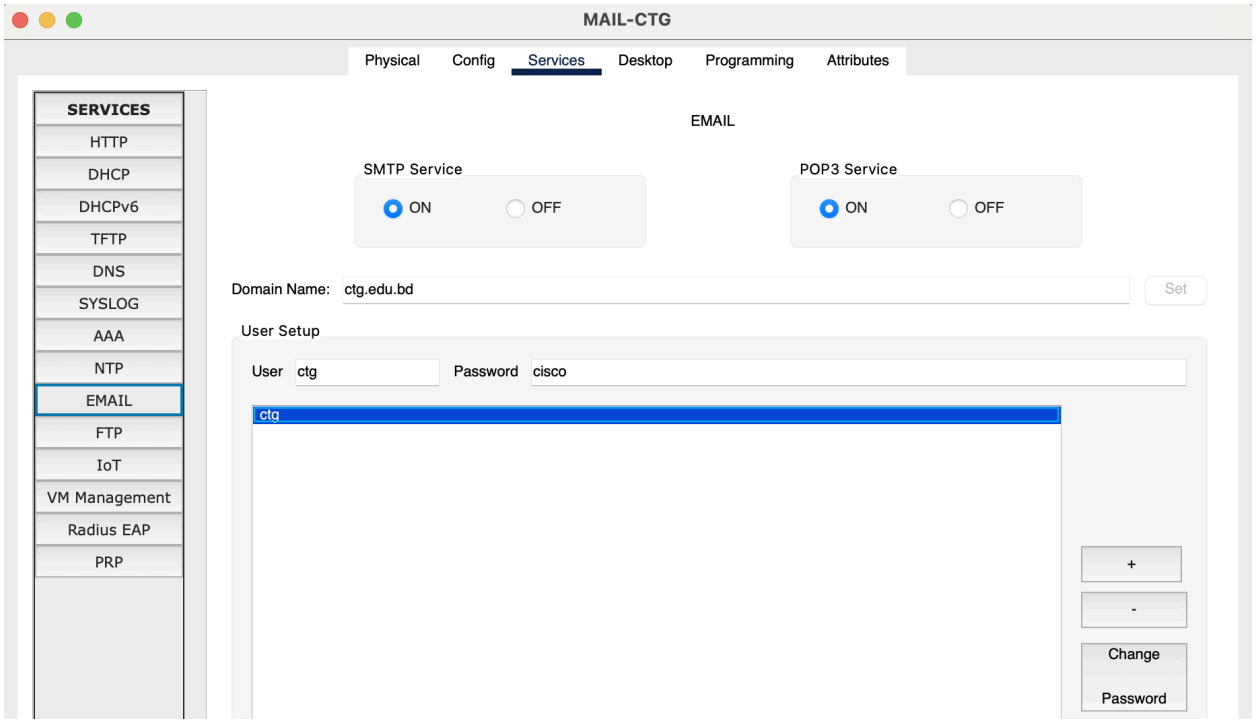
Fig: WEB-CTG can be accessed from PC-CTG2

7. Email server configuration details:

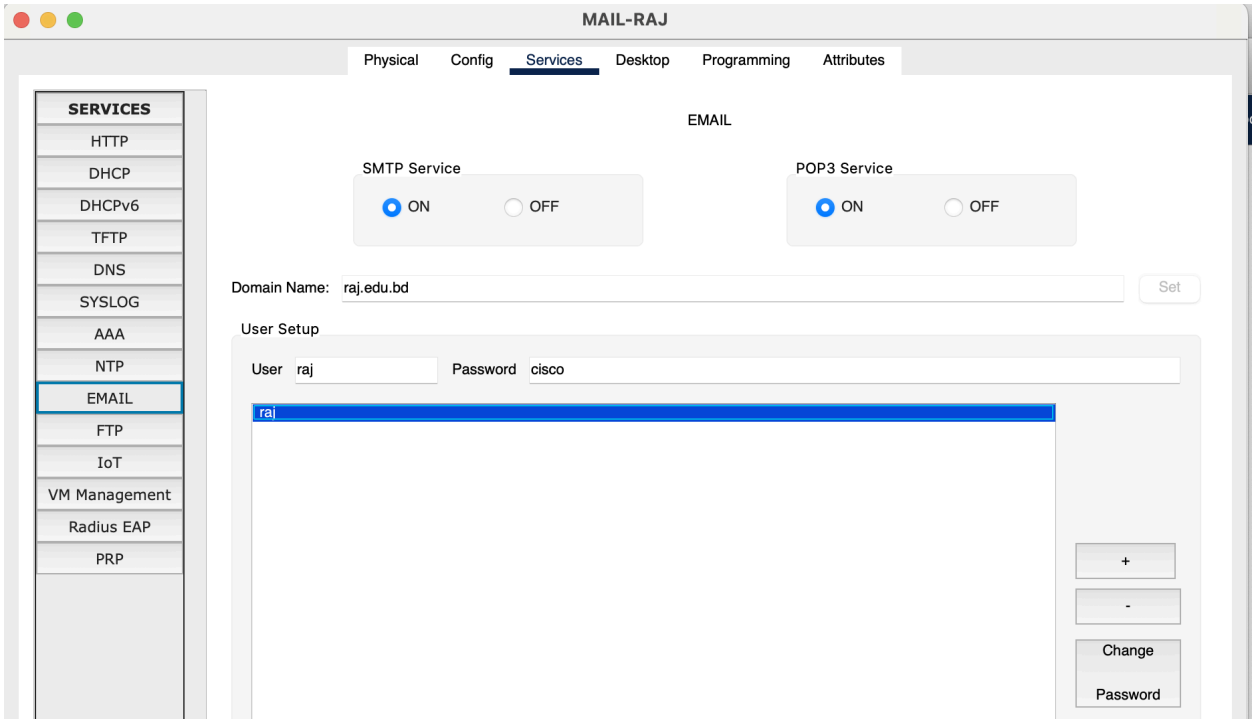
MAIL-DHK:



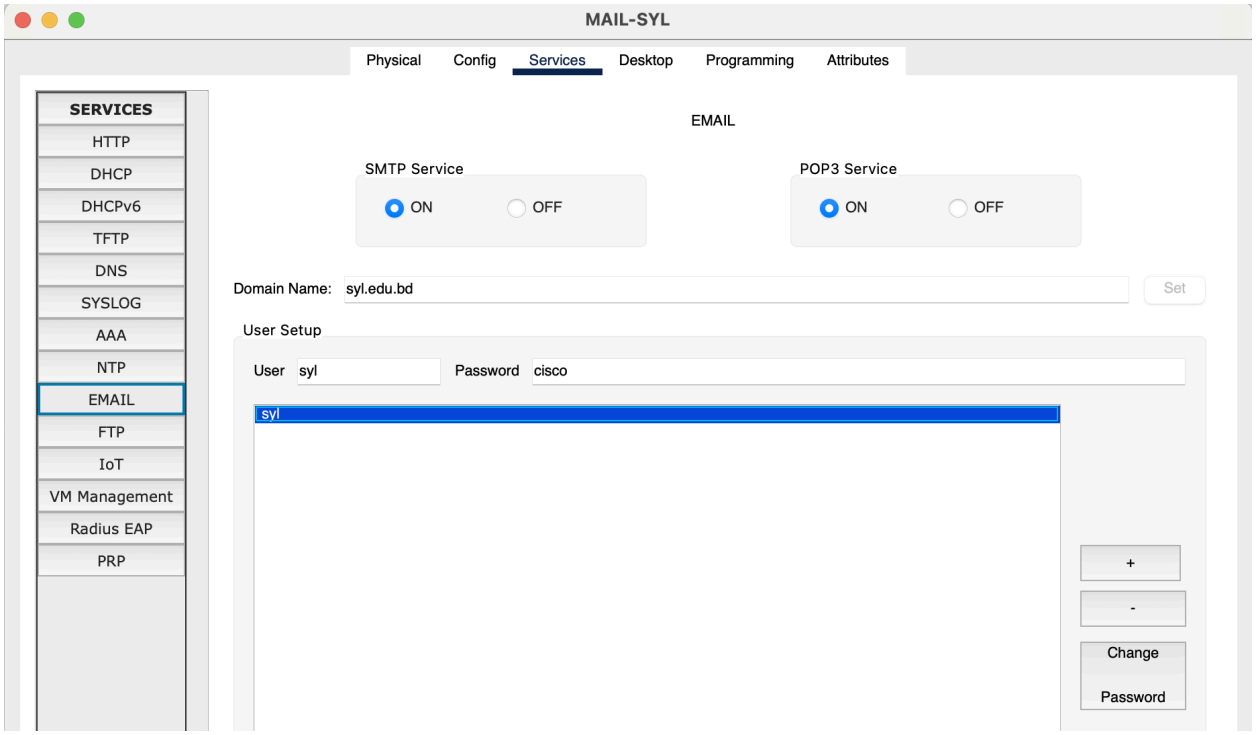
MAIL-CTG:



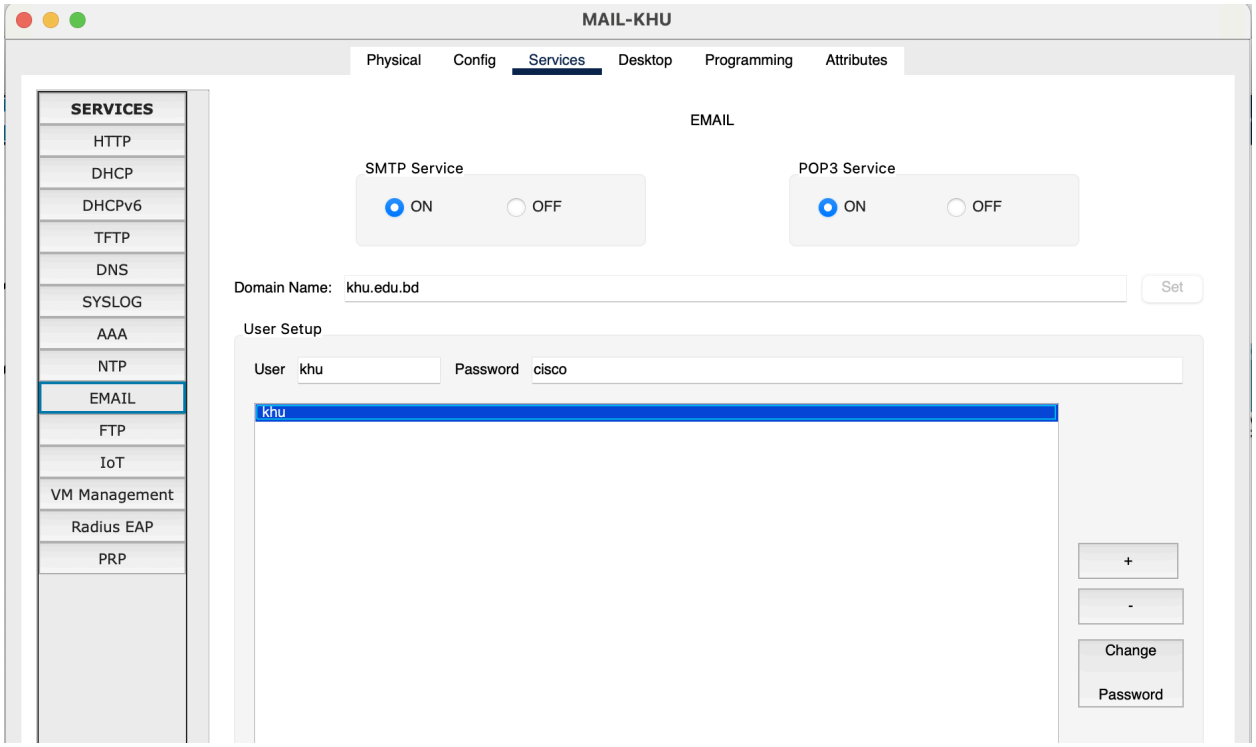
MAIL-RAJ:



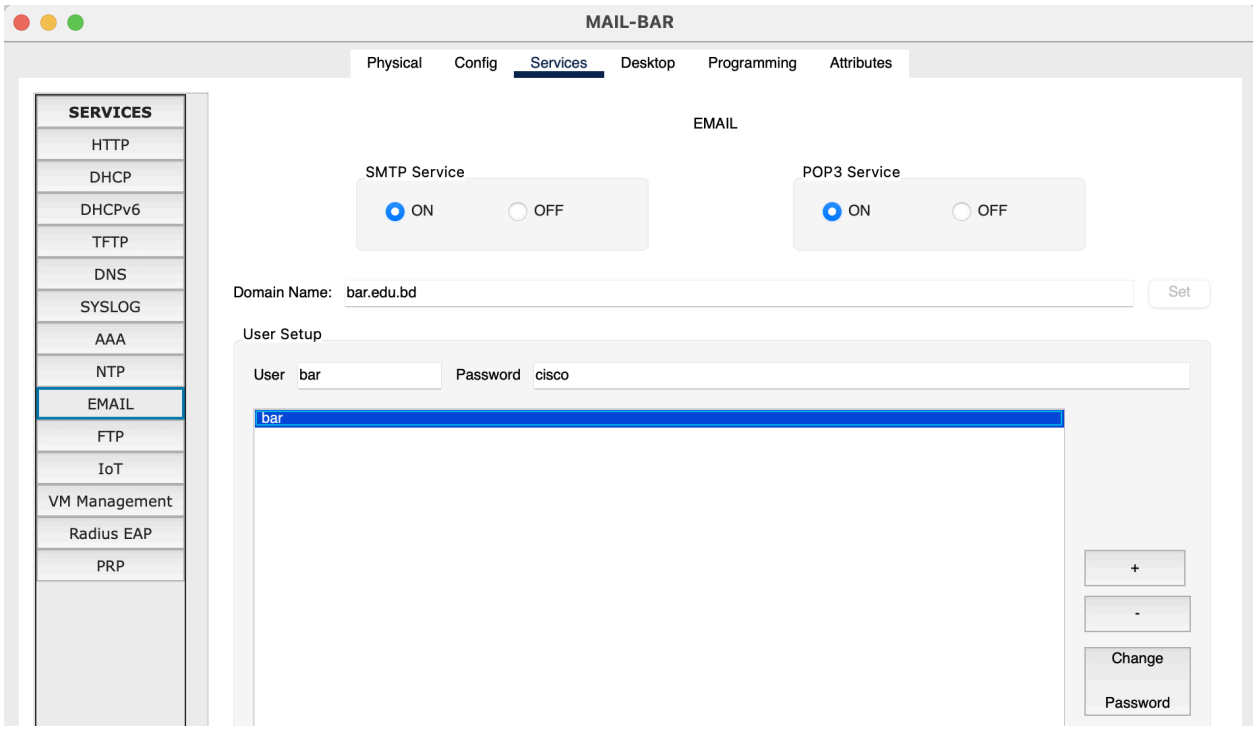
MAIL-SYL:



MAIL-KHU:



MAIL-BAR:



Cross-Board Mail Check:

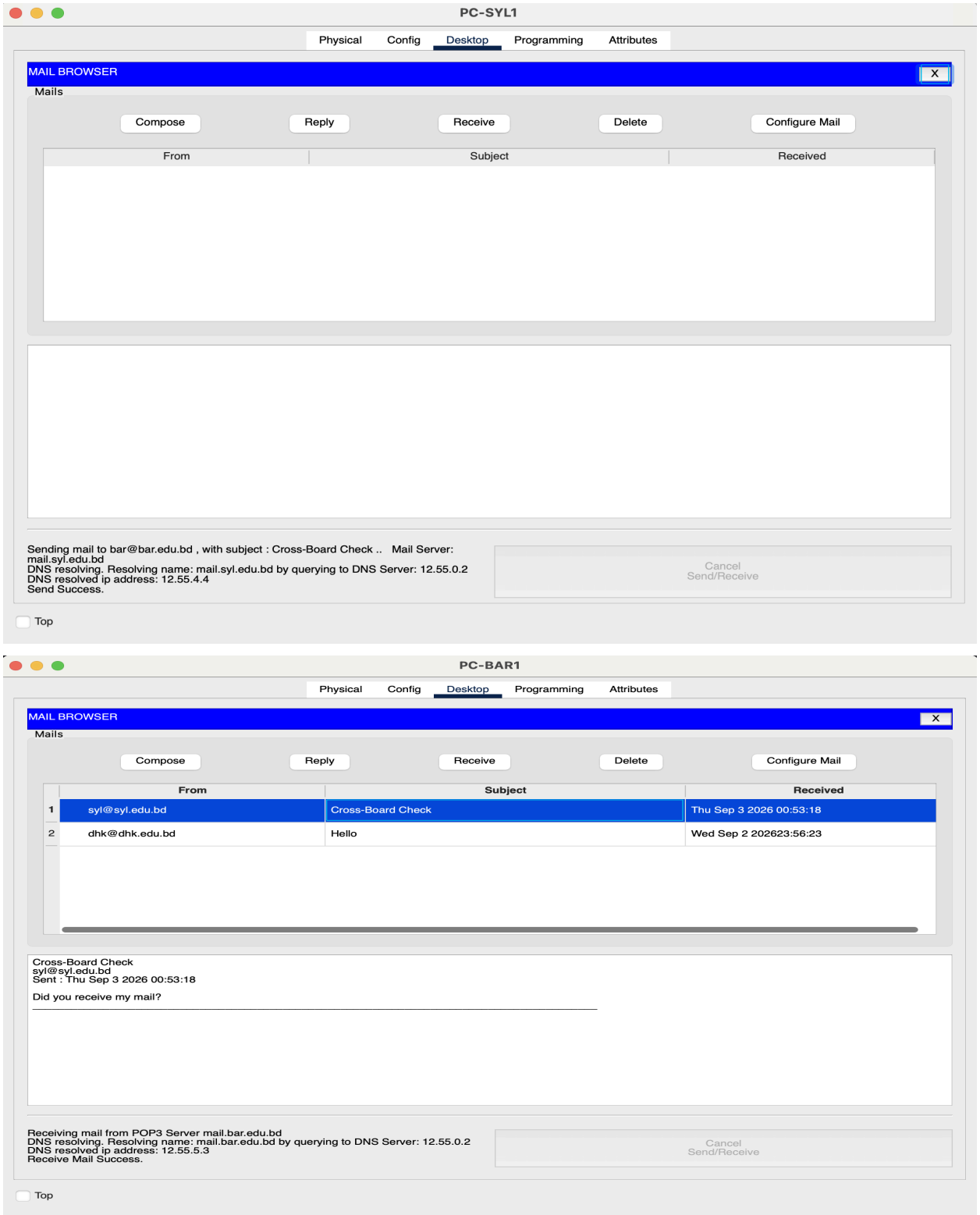


Fig: Mail sent from SYL Board (PC-SYL1) to BAR Board (PC-BAR1)

8. Assumptions:

- Router platform — Cisco 2911 (ISR G2) with HWIC-2T serial modules, two on DHK and one each on CTG, KHU and BAR. Chosen because DHK requires five interfaces, which exceeds what a single-module router provides.
- Clock rate 64000 on all DCE ends — serial links in Packet Tracer stay down without a clock rate on the DCE side. The specific value is arbitrary; only its presence matters.
- Two PCs per board represent all hosts — stated in the brief. The subnets are still sized for the full stated host counts, not for two devices.
- Explicit statics on KHU and BAR rather than a default route — both would function; explicit routes were used because the brief says those routers must know about all other networks.
- Bare-domain DNS A records — not listed in the brief, but required in practice for Packet Tracer's SMTP to deliver mail across domains.
- Fa0/24 reserved for the router uplink on every LAN switch — an organisational choice that keeps Fa0/1 onward free for hosts and makes screenshots consistent.